Streamlining IT Procurement:
Automating Standard Laptop Orders with WI

A Final Year Project Document – Problem-Solution Fit & System Overview

Date	15 March 2026
Team ID	IT-PRJ-2026-07
Project Name	Streamlining IT Procurement: Automating standard laptop orders with WI
Maximum Marks	5 Marks (Problem-Solution Fit component)

1. Abstract

This project focuses on improving the way IT departments order standard laptops inside an organisation. Currently, employees fill paper forms or send long email chains for every laptop request. This causes delays, missing approvals, budget mistakes and wrong device configurations. Our solution uses a simple Workflow Integration (WI) system that automatically handles the full process: employee request, manager approval, budget check, vendor selection, inventory update and order confirmation. The system reduces manual work, cuts the average order time from weeks to a few days and keeps clear records for audit. This document explains the problem, the existing manual process, the proposed automated system, its step-by-step workflow, benefits and future improvements.

2. Introduction

Almost every company needs to give laptops to new employees or replace old machines. In many organisations the procurement process is still done by hand. An employee raises a request, the manager signs a paper or replies to an email, the finance team checks the budget, the IT team selects a vendor and finally the order is placed. Each step waits for the next person and information is often lost or delayed.

This project introduces an automated workflow (called WI – Workflow Integration) specially designed for standard laptop orders. The goal is to keep the same approval rules that the company already uses, but to run them inside a digital system so that the process becomes faster, more transparent and less error-prone. We concentrate only on standard laptop models that are already approved by the IT department. Special or high-end machines are left outside the scope so that the system stays simple.

3. Problem Statement

The main problem is that the current laptop procurement process is slow, unclear and full of manual steps. Typical issues observed in many medium-size companies are:

- Employees do not know which form or email template to use, so requests are incomplete.
- Managers receive requests by email and sometimes forget to reply, causing long waiting times.
- Budget checks are done after the request has already moved forward, leading to rejected orders late in the process.
- IT staff spend hours chasing approvals and checking inventory by hand.
- There is no single place to see the status of an order; people keep asking “Where is my laptop?”
- Vendor selection is often based on personal preference rather than current price or stock.
- Paper records or scattered emails make it hard to prepare reports for audit or management.

Because of these problems the average time from request to laptop delivery is often 15–25 working days. This affects employee productivity and creates unnecessary work for the IT and procurement teams.

4. Objectives

The project aims to achieve the following clear objectives:

1. Design a simple digital form for employees to raise a standard laptop procurement request.
2. Build an automated workflow that routes the request to the correct manager for approval.
3. Add an automatic budget check against the department's remaining IT budget before the order moves forward.
4. Integrate a small vendor selection step that picks the preferred supplier based on price and available stock.
5. Update the internal inventory record once the order is confirmed.
6. Provide a dashboard where the requester, manager and IT team can see the current status of every request.
7. Reduce the end-to-end processing time of a standard laptop order to less than 5 working days.
8. Keep a complete digital audit trail of every approval and decision.

5. Existing System

In the present system the process is completely manual or only partly supported by email. A typical sequence looks like this:

1. Employee downloads a Word or Excel form (or writes a free-text email) and fills in laptop model, justification and required date.
2. The form is sent by email or handed as a paper copy to the reporting manager.
3. Manager reviews and either signs or replies with "Approved" or "Rejected".
4. If approved, the request is forwarded to the IT procurement officer.
5. Procurement officer checks the remaining budget with the finance team (usually by another email or phone call).
6. If budget is available, the officer selects a vendor from a small list and places a purchase order.
7. The vendor delivers the laptop; IT team receives it, updates a spreadsheet inventory and informs the employee.

Problems of the existing system include long waiting times, missing documents, no automatic status updates, and difficulty in preparing monthly procurement reports. There is also a high chance of human error when copying data from one email to another or from paper to spreadsheet.

6. Proposed System

The proposed system replaces the manual chain with a single Workflow Integration (WI) platform. All steps stay the same in meaning, but they are executed inside a digital workflow engine. Key features of the proposed system are:

• A web-based request form with drop-down lists for approved laptop models only. Employees cannot select non-standard machines.

• Automatic routing of the request to the correct manager based on the employee's department and hierarchy data.

• Manager receives a notification (email + dashboard alert) and can approve or reject with one click. Comments are stored.

- After manager approval the system automatically queries the budget service or a shared budget table to verify remaining funds.
- If the budget check passes, the system presents a short vendor list (pre-configured preferred suppliers) and allows the procurement officer to confirm the final vendor and price.
- Once the purchase order is generated, the system updates the inventory module (or a central inventory spreadsheet via API) so that stock levels stay accurate.
- Every stakeholder can open a personal dashboard and see the exact stage of the request (Submitted / Waiting Manager / Budget Check / Vendor Selection / Ordered / Delivered).
- All actions are logged with user ID, timestamp and decision, creating a ready-to-use audit trail.

The system is designed to be lightweight. It can be built using a low-code workflow tool or a simple web application with a relational database. No heavy ERP is required for the first version.

7. System Architecture & Workflow

The overall architecture consists of four main layers:

1. **Presentation Layer** – Web forms and dashboards used by employees, managers and IT staff.
2. **Workflow Engine** – The core WI component that moves the request from one stage to the next according to defined rules.
3. **Business Services** – Budget check service, vendor catalogue, inventory update service and notification service.
4. **Data Layer** – Database tables that store requests, approvals, budget balances, vendor master and inventory records.

Below is the step-by-step automated laptop order workflow implemented by the system:

Step 1 – Employee raises Procurement Request

The employee logs into the WI portal, selects “New Laptop Request”, chooses a standard model from the approved list, enters justification and required delivery date, then submits. The system creates a unique request ID and stores the data.

Step 2 – Manager Approval

The workflow engine looks up the employee’s reporting manager and sends a notification. The manager opens the request, reviews the justification and either Approves or Rejects. If rejected, the employee receives the reason and the process stops.

Step 3 – Automatic Budget Check

After approval the system reads the department’s remaining IT budget (from a budget table or external finance API). If the laptop cost is within the remaining budget, the request moves forward. If not, the request is marked “Budget Insufficient” and both the manager and employee are informed.

Step 4 – Vendor Selection

The procurement officer (or an automated rule) sees a short list of preferred vendors for that laptop model together with current price and estimated delivery days. The officer confirms the vendor. The system then generates a draft purchase order.

Step 5 – Inventory & Order Confirmation

Once the purchase order is confirmed, the system updates the inventory record (expected stock increase) and marks the request as “Ordered”. A confirmation email is sent to the employee with the expected delivery date.

Step 6 – Delivery & Closure

When the laptop arrives, the IT team marks the request as “Delivered” and the inventory is adjusted to actual stock. The workflow is closed and all logs remain available for audit.

8. Benefits

- Faster processing – average cycle time drops from 15–25 days to 3–5 days for standard laptops.
- Fewer errors – mandatory fields and drop-downs prevent incomplete requests and wrong models.
- Clear visibility – every stakeholder sees the live status without sending “any update?” emails.
- Better budget control – budget is checked automatically before the order is placed.
- Consistent vendor choice – preferred suppliers are suggested based on price and stock, reducing ad-hoc decisions.
- Complete audit trail – every approval, rejection and budget check is recorded with timestamp and user.
- Less manual work for IT and procurement staff – they only handle exceptions instead of chasing every request.
- Improved employee experience – new joiners or staff needing a replacement receive their laptop sooner.

9. Conclusion

The current manual way of ordering standard laptops creates delays, confusion and extra work for everyone involved. By introducing a Workflow Integration (WI) system that automates the procurement request, manager approval, budget check, vendor selection and inventory update, the organisation can make the process much faster and more reliable. The proposed system keeps the same business rules that managers and finance already use, but executes them inside a digital workflow. This project shows that even a focused automation of one common IT procurement activity can deliver clear time savings and better control. The design is kept simple so that it can be implemented by a small team and later extended to other device types if needed.

10. Future Scope

Several improvements can be added after the first version is running successfully:

- Extend the same workflow to monitors, docking stations and other standard IT accessories.
- Connect the budget check directly to the company’s finance ERP instead of a local table.
- Add a mobile-friendly interface so managers can approve requests from their phones.
- Introduce simple AI rules that suggest the most cost-effective vendor automatically based on historical prices.
- Generate monthly analytics dashboards showing number of requests, average cycle time and budget utilisation by department.
- Integrate with the HR system so that laptop requests for new joiners are created automatically when an offer letter is accepted.
- Support multi-level approval for high-value or non-standard requests while keeping the standard path fully automated.

11. References

1. Office of Government Commerce (UK). (2008). Procurement and Contract Management Guidance. London: OGC.

2. Laudon, K. C., & Laudon, J. P. (2020). Management Information Systems: Managing the Digital Firm (16th ed.). Pearson.

3. Workflow Management Coalition. (2019). Workflow Reference Model – Concepts and Terminology. Available at: https://www.wfmc.org

4. ITIL Foundation. (2019). ITIL 4 Foundation Publication. Axelos Limited.

5. Chopra, S., & Meindl, P. (2019). Supply Chain Management: Strategy, Planning and Operation (7th ed.). Pearson.

6. Internal company procurement policy documents and standard operating procedures (used for understanding current approval rules).

7. Low-code platform documentation (example: Microsoft Power Automate / Appian / similar tools) for workflow design patterns.

End of Document – Streamlining IT Procurement: Automating standard laptop orders with WI
Prepared as original student project work – March 2026